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Further range expansion of the blue crab *Callinectes sapidus* (Rathbun, 1896) (Crustacea: Decapoda: Brachyura) in Turkish waters, Northern Aegean Sea: insight into distribution depth

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Abstract: A new record of *Callinectes sapidus* from the northern eastern part of the Aegean Sea is presented, based on a live single female sampled off northern part of the Gökçeada (Imbroz) Island. Surprisingly, the individual which is known shelf-estuarine species was sampled at 220 m depth with a bottom trawl on November 15th, 2015.

Résumé : *Extension de l'aire de répartition du crabe bleu* Callinectes sapidus *(Rathbun, 1896) (Crustacea : decapoda : Brachyura) dans les eaux turques, nord de la Mer Egée : une nouvelle donnée sur sa distribution verticale.* Un spécimen femelle de *Callinectes sapidus* est signalé pour le première fois dans les eaux turques au nord de la l'ile Gökçeada (Imbroz). Curieusement, alors que l'espèce est connue pour vivre en milieu côtier et estuarien, cet individu a été capturé par 220 m de fond le 15 November 2015.

Keywords: Callinectes sapidus ● Blue crab ● Northeastern Agean Sea

Introduction

Imbroz which is the biggest island of the Aegean Sea is located in north Aegean Sea (25°40'-26°02'E & 40°05'-40°14'N) (Fig. 1). Northeastern Aegean Sea is thought more productive than the southern part of the Aegean Sea and one of the most important fishing area of the Mediterranean (Stergiou et al., 1997; Isari et al., 2006). It is stemmed from the influence of Black Sea water outflow from the Dardanelles to the Aegean Sea. Imbroz is on dead centre of this special ground and crucial fishing area especially for trawl and longline fisheries. On the other hand, the geographical extension of Indo-Pacific species from Levantine coasts to the Aegean Sea via Mediterranean is influence the species biodiversity of this special area.

Callinectes sapidus, an invasive species in the Mediterranean, is native to the western Atlantic Ocean from Masschusetts, U.S.A., to Argentina (Williams, 1974). Recently, a climate-driven range expansion has occurred extending its northern range to Nova Scotia in the western

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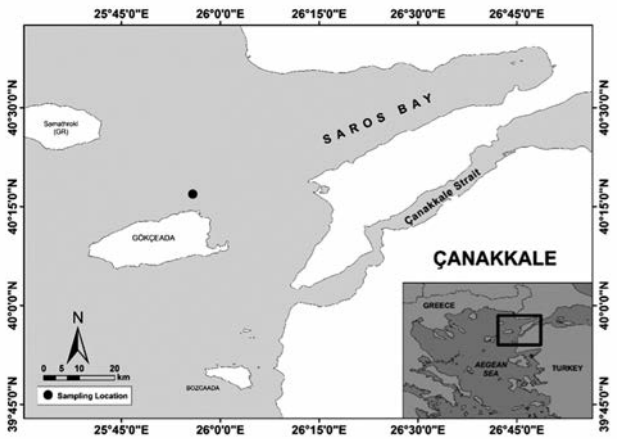


Figure 1. *Callinectes sapidus* . Map reporting the present record in Imbroz Island, North Aegean Sea.

Atlantic (Johnson, 2015). It was reported in Bay of Biscay (Vincent, 1986), Belgium and Netherlands (ICES WGITMO, 2001 & 2004). Considering the Mediterranean basin, lots of records were shown in northern Adriatic (Dulcic & Dragicevic, 2010; Manfrin et al., 2015), Cantabrian Sea (Cabal et al., 2006). Perdicaris et al. (2012) recorded specimens in Voda estuary, NW Greece (Thessalou-Legaki et al., 2012). It was also recorded from south Aegean coast of Turkey (Sumer et al., 2013), Black Sea (Yaglioglu et al., 2014; Ak et al., 2015) and Marmara Sea (Zaitsev & Öztürk, 2001). In the adjacent seas, there are two important records associated with this study. Tuncer & Bilgin (2008) recorded individual from Dardenelles and Thessalou-Legaki & Pafilis (2012) observed specimens from Chios Island (Katsanevakis et al., 2014). Here, the first record of *C. sapidus* from northeast of the Aegean Sea is presented and discussed.

Material and Methods

On 15 November 2015, one female individual was captured with a local bottom trawl vessel at a depth of 220 m, offshore the north part of the Imbroz (40°16'26"N-25°49'92"E) between the cape of Pirinç and Kaşkaval. Identification of the individual was based on abdomen morphology and coloration, according to Williams (1974). The specimen was photographed (Fig. 2), fixed in 5% formalin and later transferred to 70% ethanol, and housed in the Piri Reis Marine Museum of Canakkale Onsekiz Mart University, Canakkale (PRM-DEC / 2015 - 002).

Results and Discussion

The blue crab was captured alive and all the parts of the body maintain its characteristics as captured at 220 m depth.



Figure 2. *Callinectes sapidus*, 125.6 mm CW, from Imbroz (PRM-DEC / 2015 - 002). **A:** ventral view, **B:** dorsal view (Photograph by S. Tuncer).

The sex of the individual was observed as female from the abdomen morphology and coloration of cheliped. The specimen did not carry eggs on the abdomen. The carapace length (CL), width (CW) and individual wet weight were measured as 61 mm, 125.6 mm and 107.1 g, respectively. The left maxiliped length and depth were measured 5.50 and 1.40 mm, the right maxiliped length and width were measured 5.60 and 1.50 mm, respectively (Fig 1). The blue crab is an estuarine-dependent species and estuaries are favorable grounds for development of juveniles. Afterwards the immature females migrates to lower salinity waters to seek out mature males for mating. After mating, females migrates to the back to the estuary (Millikin & Williams, 1984). The blue crab is a shelf-estuarine species distributed at depths between 0-90 m on muddy and sandy bottoms (FAO, 2015). Conversely, in this study, the blue crab was captured at 220 m depth of northern Imbroz (Table 1). This could be the deepest distribution record of this species. Adult blue crabs are strong swimmers (Hench et al., 2004). This situation may verified of the spawning migration theory between lagoon and less saline environment. Katsanevakis & Tsiamis (2009) suggested that the blue crab presence on the island is limited due to estuarine habitat

Table 1. *Callinectes sapidus* . Sampling depths and deepest occurrence in different locations.

Author	Country	Area	Fishing Type	Habitat	Sampling Depth
Aguilar et al., 2005	United States	Chesapeake Bay	-	Brackish waters	1.2-21.3 m
Nehring & van der Meer, 2010	Germany	German Wadden Sea	Shrimp beam trawl	Marine	10 m
Dulcic & Dragicevic, 2010	Crotia	Neretva River	Hand	Freshwater	4 m
Castriota et al., 2012	Italy	Abruzzi, Central Adriatic	Gill nets	Marine	10 m
Thessalou-Legaki & Pafilis, 2012	Greece	Chios Island	Fishing nets	Brackish waters	0.4 m
Ak et al., 2015	Turkey	Eastern Black Sea	Gill nets	Marine	8-12 m
Yaglioglu et al., 2014	Turkey	Western Black Sea	Gill nets	Marine	14.5-18 m
Tuncer & Bilgin, 2008	Turkey	Dardanelles Strait	Gill nets	Marine	10 m
FAO, 2015					0-90 m
This study	**Turkey**	**North Aegean Sea**	**Bottom trawl**	**Marine**	**220 m**

preference. Thessalou-Legaki & Pafilis (2014), who first recorded the blue crab in Chios Island, revealed that urgent detailed records needed from Aegean Islands to be able to understand the habitat preference of this species (Katsanevakis et al., 2014). This record may contribute to understand complex life cycle and distribution of the blue crab. Add to this the impacts of this species insufficiently known on the native species (Zaitsev & Ozturk, 2001).

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References

Aguilar R., Hines A.H., Wolcott T.G., Wolcott D.G., Kramer M.A. & Lipcius R.N. 2005. The timing and route of movement and migration of post-copulatory female blue crabs, *Callinectes sapidus* Rathbun, from the upper Chesapeake Bay. *Journal of Experimental Marine Biology and Ecology*, **319**: 117-128.

Ak O., Hasimoglu A. & Bayram K. 2015. Southeastward expansion of the blue crab *Callinectes sapidus* (Rathbun, 1896) in the Black Sea. *Cahiers de Biologie Marine*, **56**: 397-399.

Cabal J., Antonio J., Millán P. & Carlos Arronte J. 2006. A new record of *Callinectes sapidus* Rathbun, 1896 (Crustacea: Decapoda: Brachyura) from the Cantabrian Sea, Bay of Biscay, Spain. *Aquatic Invasions*, **3**: 186-187.

Castriota L., Andaloro F., Costantini R. & Ascentiis A.D. 2012. First record of the Atlantic crab *Callinectes sapidus* Rathbun, 1896 (Crustacea: Brachyura: Portunidae) in Abruzzi waters, central Adriatic Sea. *Acta Adriatica*, **53**: 467-471.

Dulcic J. & Dragicevic B. 2010. New Record of the blue crab, *Callinectes sapidus* Rathbun, 1896, (Decapoda: Brachyura) in the Adriatic Sea. *Annales Séries Historia Naturalis*, **20**: 23-28.

FAO 2015. *Callinectes sapidus*. http://www.fao.org/fishery/species/2632/en (Accessed 20 November 2015).

Hench L.J., Forward B.R., Carr D.S., Rittschof D. & Luettich A.R. 2004. Testing a selective tidal-stream transport model: Observations of female blue crab (*Callinectes sapidus*) vertical migration during the spawning season. *Limnology & Oceanogr*aphy, **49**: 1857-1870.

ICES WGITMO 2001. Report of the working group on introductions and transfers of marine organisms, Barcelona, Spain, 21-23 March 2001. ICES CM 2001 /ACME **08**, 104 pp. available at http://www.ices.dk/iceswork/wgdetailacme.asp?wg=WGIT MO.

ICES WGITMO 2004. Report of the Working group on Introductions and Transfers of marine Organisms, Cesenatico, Italy, 25-26 March 2004. ICES CM 2004/ACME **05**, 151 pp. available at http://www.ices.dk/iceswork/wgdetailacme.asp?wg=WGIT MO.

Isari S., Ramfos A., Somarakis S., Koutsikopoulos C., Kallianiotis A. & Fragopoulu N. 2006. Mesozooplankton distribution in relation to hydrology of the Norteastern Aegean Sea, Eastern Mediterranean. *Journal of Plankton Resarch*, **28**: 241-255.

Johnson D.S. 2015. The savory swimmer swims north: a northern range extension of the blue crab *Callinectes sapidus*? *Journal of Crustacean Biology,* **35**: 105-110.

Katsanevakis S. & Tsiamis K. 2009. Records of alien marine species in the shallow coastal waters of Chios Island. *Mediterranean Marine Science*, **10**: 99-107.

Katsanevakis S., Acar Ü., Ammar I., Balcı B.A., Bekas P., Belmonte M., Chıntıroglou C.C., Consolı P., Dımıza M., Fryganıotıs K., Gerovasileiou V., Gnıscı V., Gülşahin N., Hoffman R., Issaris Y., Izquierdo-Gomez D., Izquierdo-Muñoz A., Kavadas S., Koehler L., Konstantinidis E., Mazza G., Nowell G., Önal U., Özen M.R., Pafilis P., Pastore M., Perdikaris C., Poursanidis D., Prato E., Russo F., Sicuro B., Tarkan A.N., Thessalou-Legaki M., Tiralongo F., Triantaphyllou M., Tsiamis K., Tunçer S., Turan C., Türker A. & Yapici S. 2014. New Mediterranean Biodiversity Records (October, 2014). *Mediterranean Marine Science*, **15**: 675-695.

Manfrin C., Turolla E., Sook Chung J. & Giulianni G.P. 2015. First occurrence of *Callinectes sapidus* (Rathbun, 1896) within the Sacca di Goro (Italy) and surroundings. *Notes on*

Geographic Distribution, Check List, **11**:1640.

Millikin M.R. & Williams A.B. 1984. Synopsis of biological data on the blue crab, *Callinectes sapidus* Rathbun. *NOM Technical Report*, NMFS 1, FAO Fisheries Synopsis N° **138**.

Nehring S. & van der Meer U. 2010. First record of a fertilized female blue crab, *Callinectes sapidus* Rathbun, 1896 (Crustacea: Decapoda: Brachyura), from the German Wadden Sea and subsequent secondary prevention measures. *Aquatic Invasions*, **5**: 215-218.

Sumer C., Teksam I., Karatas H., Beyhan T. & Aylin C.M. 2013. Growth and reproductive biology of the blue crab, *Callinectes sapidus* Rathbun. 1986, in the Beymelek Lagoon (Southwestern Coast of Turkey). *Turkish Journal of Fisheries and Aquatic Sciences*, **13**: 675-684.

Stergiou K.I., Christou E.D., Georgopoulos D., Zenetos A. & Souvermezoglou C. 1997. The hellenic seas: physics, chemistry, biology and fisheries. *Oceanography and Marine Biology - An Annual Review*, **35**: 415-538.

Thessalou-Legaki M., Aydogan O., Bekas P., Bilge G., Boyacı Y.O., Brunelli E., Circosta V., Crocetta F., Durucan F., Erdem M., Ergalavou A., Filiz H., Fois F., Gouva E., Kapiris K., Katsanevakis S., Kljajic Z., Konstantinidis E., Konstantinou G., Koutsogiannopoulos D., Lamon S., Macic V., Mazzette R., Meloni D., Mureddu A., Paschos I., Perdicaris C., Piras F., Poursanidis D., Ramos-Espla A.A., Rosso A., Sordino P., Sperone E., Strerioti A., Taskin E., Toscano F., Tripepi S., Tsiakkiros L. & Zenetos A. 2012. New Mediterranean Biodiversity Records (December 2012) *Mediterranean Marine Science*, **13/2**: 312-327.

Tuncer S. & Bilgin S. 2008. First record of *Callinectes sapidus* Rathbun, 1896 (Crustacea: Decapoda: Brachyura) in the Dardanelles, Canakkale, Turkey. *Aquatic Invasions*, **3**: 469.

Vincent T. 1986. Les captures de *Callinectes sapidus* (Rathbun, 1896) en baie de Seine, entre 1975 et 1984. *Bulletin Trimestriel de la Société Géologique de Normandie et des Amis du Muséum du Havre*, **LXXIII**: 13-15.

Williams A.B. 1974. The swimming crabs of the genus *Callinectes* (Decapoda: Portunidae). *Fisheries Bulletin*, **72**: 685-798.

Yaglioglu D., Turan C. & Ogreden T. 2014. First record of blue crab *Callinectes sapidus* (Rathbun, 1896) (Crustacea, Brachyura, Portinidae) from the Turkish Black Sea coast. *Journal of Black Sea/Mediterranean Environment*, **20**: 13-17.

Zaitsev Y. & Ozturk B. 2001. *Exotic species in the Aegean, Marmara, Black, Azov and Caspian Seas.* Turkish Marine Research Foundation: Istanbul, Turkey, 267 pp.